CALCULUS & LINEAR ALGEBRA

Quick Review • Definitions • Core Formulae • Essential Graphs

Based on the MATH 2183 course outline
United International University • Summer 2026

Reference direction: Howard Anton — Calculus and Elementary Linear Algebra; Earl D. Rainville / George F. Simmons-style standard ODE methods as represented by the course's Ross reading; Schaum's Outline for matrix methods.

A compact study companion. The syllabus determines the topic coverage; the formulas below are standard supporting results for those topics. Check lecture notes for instructor-specific notation and examples.

How to use this guide

This summary follows the topics listed in the uploaded MATH 2183 outline. It is designed for quick revision, not as a replacement for textbook proofs, assigned examples, or class notes.

Font note: The PDF uses an embedded Unicode font for ordinary text and common mathematical symbols. Equations are typeset as readable text rather than pasted screenshots, so they remain sharp when zoomed or printed.

Course coverage at a glance

Block	Topics included in the syllabus
Calculus I	Increasing/decreasing functions; concavity; inflection points; critical and stationary points; relative extrema; polynomials; first- and second-derivative tests; absolute extrema (basic idea).
Multivariable calculus	Partial derivatives; higher-order partials; chain rule; tangent planes; maxima/minima of functions of two variables.
Differential equations	Order/degree; linear vs nonlinear; separable and first-order linear ODEs; applications; exact and homogeneous ODEs; linear homogeneous/non-homogeneous equations; higher-order constant-coefficient ODEs; undetermined coefficients.
Linear algebra	Matrix algebra and special matrices; linear systems; row operations and echelon forms; Gauss–Jordan; inverse; determinants/cofactor expansion; Cramer’s rule; rank and consistency; vector spaces; linear dependence/independence; transformations; eigenvalues/eigenvectors; diagonalization and applications.

Suggested reference books

Howard Anton, Calculus — the syllabus cites Chapter 4.1–4.2 and Chapter 13.3, 13.5, 13.7–13.8; Ross — cited for differential equations; Schaum’s Outline — cited for matrix algebra. The outline does not provide full bibliographic editions, so verify the exact edition used in class.

1. Function analysis: monotonicity, concavity and extrema

Increasing and decreasing

A function is increasing where larger x-values produce larger f(x)-values; decreasing where larger x-values produce smaller f(x)-values. Use the first derivative to identify intervals.

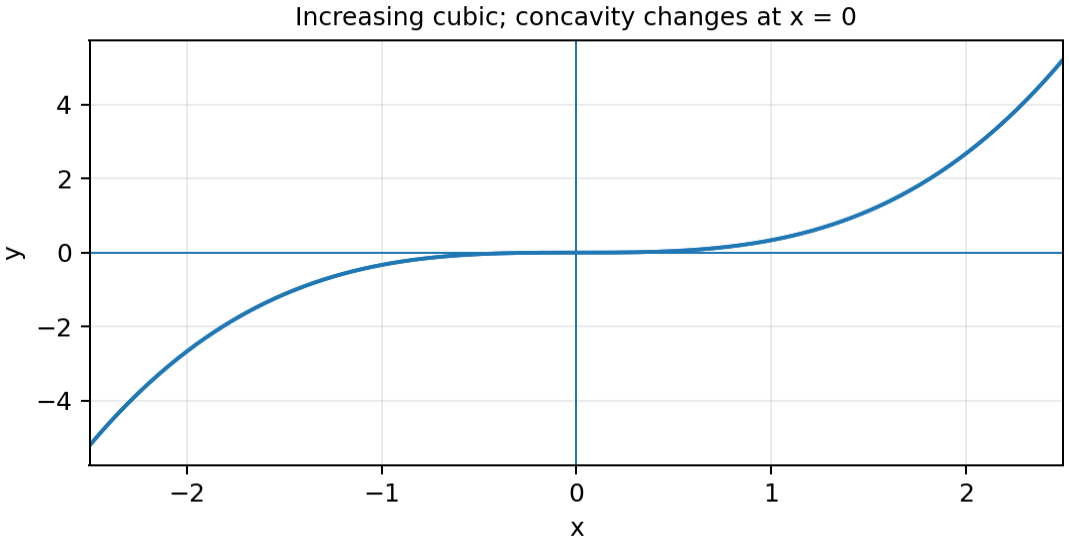
If $f'(x) > 0$ on an interval, f is increasing there. If $f'(x) < 0$, f is decreasing there.

Concavity and inflection points

Concavity describes how the slope changes. Concave up means the slopes are increasing; concave down means the slopes are decreasing. An inflection point is a point where concavity changes.

$f''(x) > 0 \Rightarrow$ concave up. $f''(x) < 0 \Rightarrow$ concave down.

Possible inflection x-values: $f''(x) = 0$ or $f''(x)$ undefined; then test whether concavity actually changes.



Critical points, stationary points and relative extrema

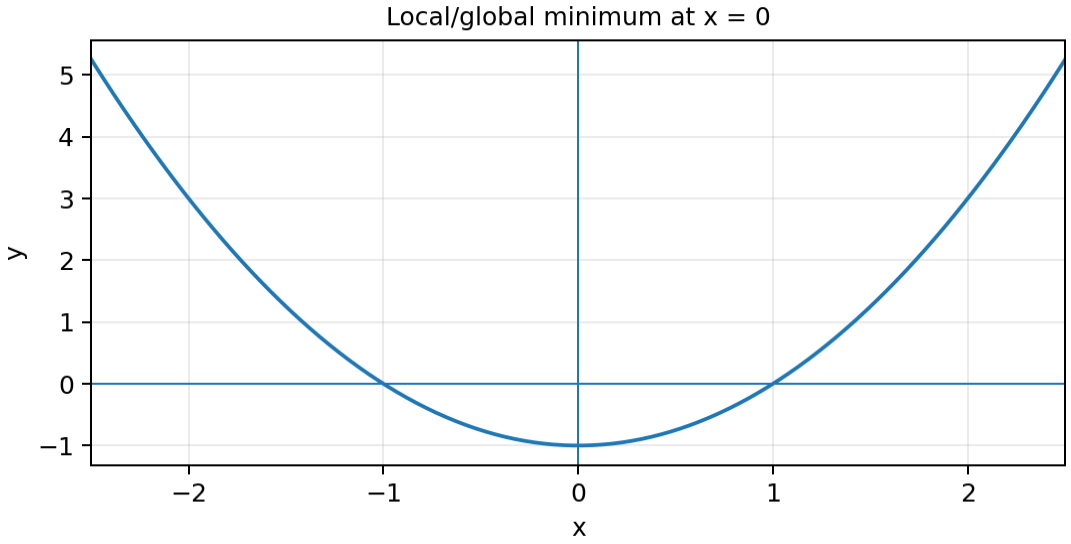
Critical number c: c is in the domain of f and $f'(c) = 0$ or $f'(c)$ does not exist.

Stationary point: $f'(c) = 0$. Every stationary point is critical, but not every critical point is stationary.

A relative (local) maximum is higher than nearby function values; a relative minimum is lower than nearby values.

First derivative test: f' changes + to − $\Rightarrow$ local maximum; − to + $\Rightarrow$ local minimum; no sign change $\Rightarrow$ neither.

Second derivative test: if $f'(c)=0$ and $f''(c)>0 \Rightarrow$ local minimum; if $f''(c)<0 \Rightarrow$ local maximum; if $f''(c)=0 \Rightarrow$ inconclusive.



Absolute maximum/minimum and polynomials

On a closed interval [a,b], a continuous function attains an absolute maximum and minimum. Compare f-values at all critical numbers inside (a,b) and at endpoints a and b. For polynomials, f is defined and continuous for every real x; end behavior is controlled by the leading term.

> Even degree: both ends point the same way. Odd degree: ends point in opposite directions. The leading coefficient determines whether the right end rises or falls.

Workflow: (1) find f′; (2) solve f′=0 and find where f′ is undefined; (3) build a sign chart; (4) use f″ for concavity; (5) evaluate candidate points and endpoints when an interval is given.

2. Partial derivatives and multivariable calculus

Partial derivatives

For f(x,y), a partial derivative differentiates with respect to one variable while treating the other as constant.

$f_x = \partial f/\partial x$; $f_y = \partial f/\partial y$.

$f_{xx} = \partial^2 f/\partial x^2$; $f_{yy} = \partial^2 f/\partial y^2$; $f_{xy} = \partial^2 f/(\partial y \partial x)$; $f_{yx} = \partial^2 f/(\partial x \partial y)$.

If the relevant second partial derivatives are continuous near a point, then $f_{xy} = f_{yx}$ (Clairaut's theorem).

Chain rule

If z=f(x,y), x=x(t), y=y(t): $dz/dt = f_x \cdot dx/dt + f_y \cdot dy/dt$.

If z=f(x,y), x=x(u,v), y=y(u,v): $z_u = f_x x_u + f_y y_u$; $z_v = f_x x_v + f_y y_v$.

Tangent plane and linearization

For z=f(x,y), the tangent plane at (a,b,f(a,b)) approximates the surface near that point.

$z - f(a,b) = f_x(a,b)(x-a) + f_y(a,b)(y-b)$.

Linearization: $L(x,y)=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$.

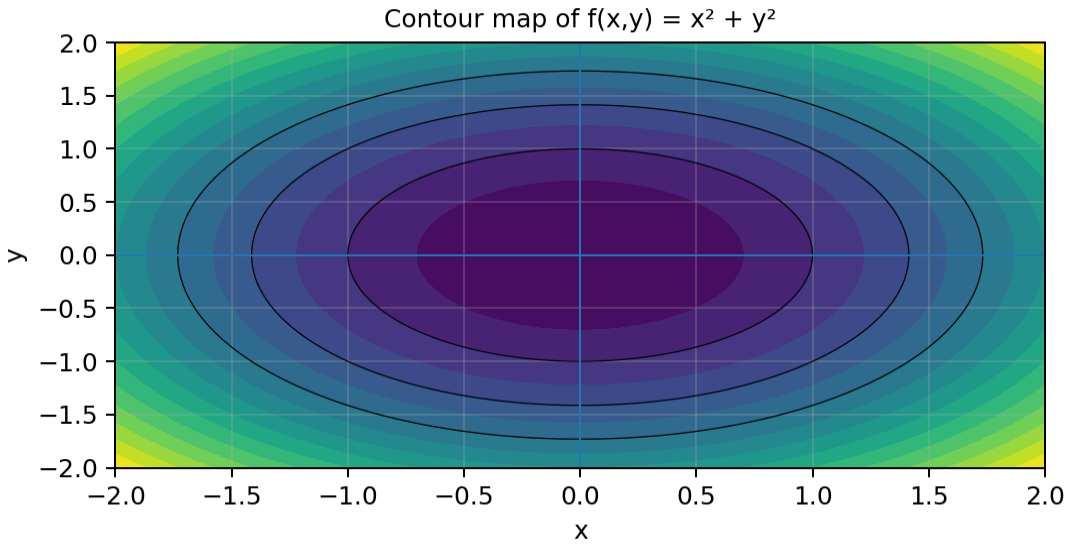
Maxima/minima of two-variable functions

Interior critical point: $f_x(a,b)=0$ and $f_y(a,b)=0$ (also check where partials fail to exist).

$D = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2$.

$D>0$ and $f_{xx}>0 \Rightarrow$ local min; $D>0$ and $f_{xx}<0 \Rightarrow$ local max; $D<0 \Rightarrow$ saddle; $D=0 \Rightarrow$ inconclusive.

For absolute extrema on a closed bounded region, examine interior critical points and the boundary. The syllabus asks for maxima/minima of two-variable functions; boundary methods may depend on class notes.



Graph reading: contour curves connect points with equal function value. For $f(x,y)=x^2+y^2$, the contours are concentric circles and the surface has a minimum at the origin.

3. Differential equations: first-order methods

Basic vocabulary

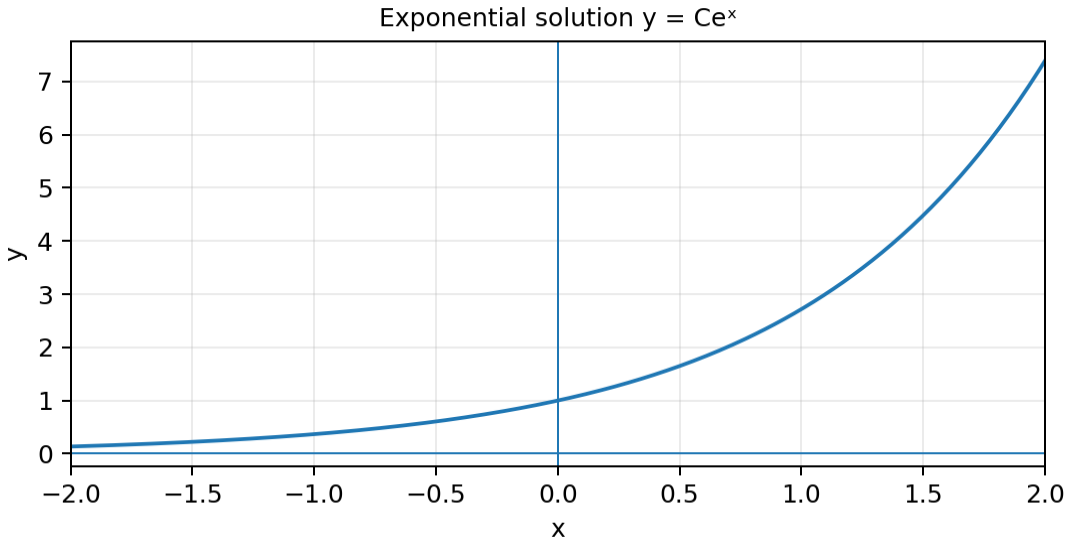
Term	Meaning
Differential equation	An equation involving an unknown function and one or more of its derivatives.
Order	The highest derivative appearing in the equation.
Degree	The power of the highest-order derivative, after the equation is polynomial in derivatives.
Linear ODE	The unknown function and its derivatives appear only to the first power, are not multiplied together, and have coefficients depending only on the independent variable.
Nonlinear ODE	An equation that fails the linearity conditions.

Separable equations

dy/dx = g(x)h(y) ⇒ dy/h(y) = g(x) dx.

Integrate both sides: ∫dy/h(y) = ∫g(x)dx + C.

Keep constant/equilibrium solutions that may be lost if dividing by h(y).



First-order linear ODE

Standard form: y′ + P(x)y = Q(x).

Integrating factor: μ(x)=e^(∫P(x)dx).

Solution: y = [∫ μ(x)Q(x) dx + C]/μ(x).

Exact ODE

M(x,y) dx + N(x,y) dy = 0 is exact if $M_y = N_x$ (on a suitable region).

Find a potential function F with F_x=M and F_y=N; the implicit solution is F(x,y)=C.

Homogeneous first-order ODE

For dy/dx = F(y/x), use y=vx, so dy/dx = v + x dv/dx. This often reduces the equation to a separable equation in v and x.

Substitution: y=vx; y′=v+xv′.

Applications

Common models include growth/decay, cooling, mixing and simple motion. First identify the dependent variable and assumptions, then form the ODE, solve it, and apply the initial condition.

Exponential growth/decay: dP/dt=kP ⇒ P(t)=P_0e^(kt).

Newton cooling: dT/dt=−k(T−T_a) ⇒ T−T_a=(T_0−T_a)e^(−kt).

4. Higher-order differential equations

Linear ODEs with constant coefficients

$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x)$, where the a_i are constants.

For the homogeneous equation, assume y=e^(rx). This gives the characteristic (auxiliary) equation.

$a_n r^n + a_{n-1} r^{n-1} + \dots + a_1 r + a_0 = 0$.

Characteristic-root cases

Roots	Complementary solution y_c
Distinct real roots $r_1,\dots,r_n$	$C_1 e^{(r_1 x)} + \dots + C_n e^{(r_n x)}$
Repeated real root r of multiplicity m	$e^{(rx)}(C_1 + C_2 x + \dots + C_m x^{(m-1)})$
Complex roots $\alpha \pm i\beta$	$e^{(\alpha x)}[C_1 \cos(\beta x) + C_2 \sin(\beta x)]$
Repeated complex roots	Multiply the corresponding sin/cos terms by 1, x, x^2, ... as required by multiplicity.

Non-homogeneous equations and undetermined coefficients

General solution: y = y_c + y_p, where y_c solves the homogeneous equation and y_p is one particular solution.

Choose a trial y_p based on g(x), substitute into the ODE and solve for unknown coefficients. If the trial duplicates a term in y_c, multiply the trial by x enough times to remove overlap.

Forcing term g(x)	Usual trial form (before resonance adjustment)
Polynomial of degree m	General polynomial of degree m
e^(ax)	A e^(ax)
sin(bx) or cos(bx)	A cos(bx)+B sin(bx)
e^(ax) times polynomial	e^(ax) × general polynomial
e^(ax)sin(bx) or e^(ax)cos(bx)	e^(ax)[A cos(bx)+B sin(bx)]

Ross reading in the outline names higher-order constant-coefficient equations (three root cases) and the method of undetermined coefficients. Follow the class notes for the exact worked examples and notation.

5. Matrices and systems of linear equations

Matrix algebra

If $A=[a_{ij}]$ is $m\times n$ and B is $n\times p$, then AB is $m\times p$ and $(AB)_{ij} = \Sigma_k\, a_{ik}b_{kj}$.

$(A+B)^T=A^T+B^T$; $(AB)^T=B^TA^T$; $(AB)^{-1}=B^{-1}A^{-1}$ when inverses exist.

Identity: $AI=IA=A$. A is symmetric if $A^T=A$. A is diagonal if all off-diagonal entries are zero.

Upper triangular matrices have zero entries below the main diagonal; lower triangular matrices have zero entries above it.

Linear systems and row operations

Write a system as $Ax=b$ or as an augmented matrix $[A|b]$. Elementary row operations preserve the solution set:

(1) swap two rows; (2) multiply a row by a nonzero constant; (3) add a multiple of one row to another.

Row echelon form (REF): nonzero rows above zero rows; each leading entry moves right as you go down. Reduced row echelon form (RREF) also has leading 1s and each pivot is the only nonzero entry in its column.

Gauss–Jordan elimination

Augment $[A|b]$, use row operations to reach RREF, then read off the solution. A row $[0\ \dots\ 0 \mid c]$ with $c\neq 0$ means inconsistent. Free variables indicate infinitely many solutions when the system is consistent.

Inverse of a square matrix

A is invertible iff $[A|I]$ can be row-reduced to $[I|A^{-1}]$.

For a 2×2 matrix $A=[[a,b],[c,d]]$, $\det(A)=ad-bc$.

$A^{-1} = (1/(ad-bc))\ [[d,-b],[-c,a]]$, provided $ad-bc \neq 0$.

Determinants, cofactor expansion and Cramer’s rule

Cofactor: $C_{ij} = (-1)^{(i+j)}\, M_{ij}$, where M_{ij} is the minor determinant after deleting row i and column j.

Expansion along row i: $\det(A)=\Sigma_j\, a_{ij}\, C_{ij}$. The same expansion works along a column.

Cramer’s rule (when $\det A\neq 0$): $x_i = \det(A_i)/\det(A)$, where A_i replaces column i of A by b.

6. Rank, consistency, vector spaces and dependence

Rank theorem / consistency test

rank(A) = number of pivot positions = dimension of the column space.

For Ax=b with n unknowns: consistent iff rank(A)=rank([A|b]).

If consistent and rank(A)=n ⇒ unique solution. If consistent and rank(A)<n ⇒ infinitely many solutions. If ranks differ ⇒ no solution.

Vector spaces and subspaces

A vector space is a set with vector addition and scalar multiplication satisfying the standard closure and algebraic axioms. A subspace is a nonempty subset closed under addition and scalar multiplication.

For vectors in R^n, the zero vector must belong to a subspace. A set of vectors is linearly independent if the only solution of $c_1v_1+\ldots+c_kv_k=0$ is all $c_i=0$.

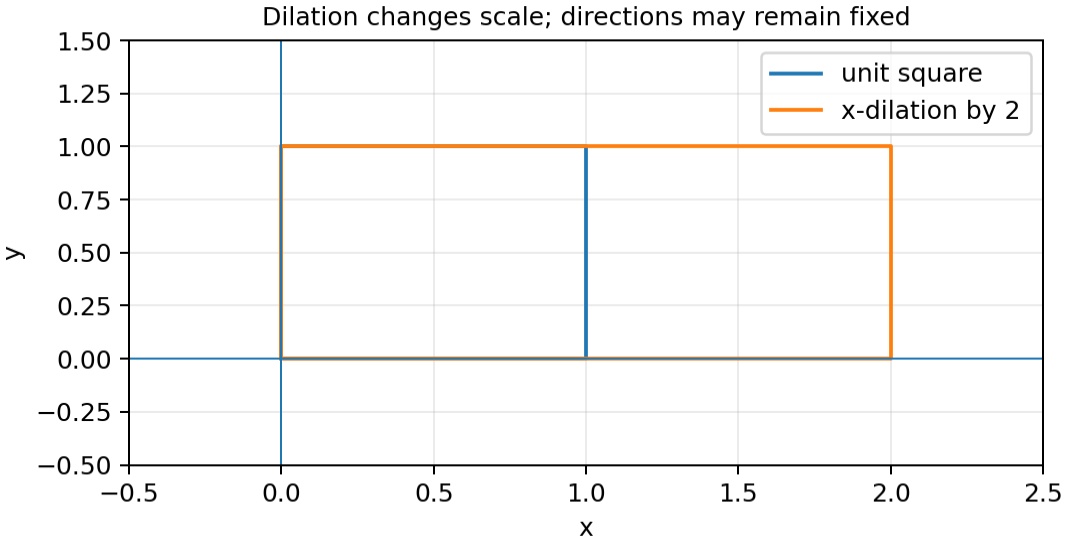
Linear dependence: there exists a nontrivial coefficient choice (not all zero) such that $c_1v_1+\ldots+c_kv_k=0$.

To test independence, place the vectors as columns of a matrix and solve Ac=0. A free variable / nontrivial nullspace means dependence.

Linear transformations: dilation and contraction

A linear transformation T satisfies T(u+v)=T(u)+T(v) and T(cv)=cT(v). In coordinates, T(x)=Ax. A dilation scales lengths; a contraction scales toward the origin by a factor between 0 and 1.

Uniform scaling by k: T(x)=kx, represented by A=kI. If |k|>1 it expands; if 0<|k|<1 it contracts.



Quick facts

Concept	Test / key fact
Column space	Span of the columns of A; its dimension is rank(A).
Null space	Solutions of Ax=0; nullity + rank = number of columns of A.
Independent n vectors in R^n	They form a basis iff the matrix with these columns is invertible.
Square matrix	Invertible iff det(A)≠0 iff rank(A)=n iff columns are independent.

7. Eigenvalues, eigenvectors and diagonalization

Eigenvalues and eigenvectors

A nonzero vector v is an eigenvector of A if multiplying by A changes only its scale, not its direction: $Av=\lambda v$. The scalar λ is the eigenvalue.

Characteristic equation: $\det(A-\lambda I)=0$.

For each eigenvalue λ, find eigenvectors by solving $(A-\lambda I)v=0$, with $v\neq 0$.

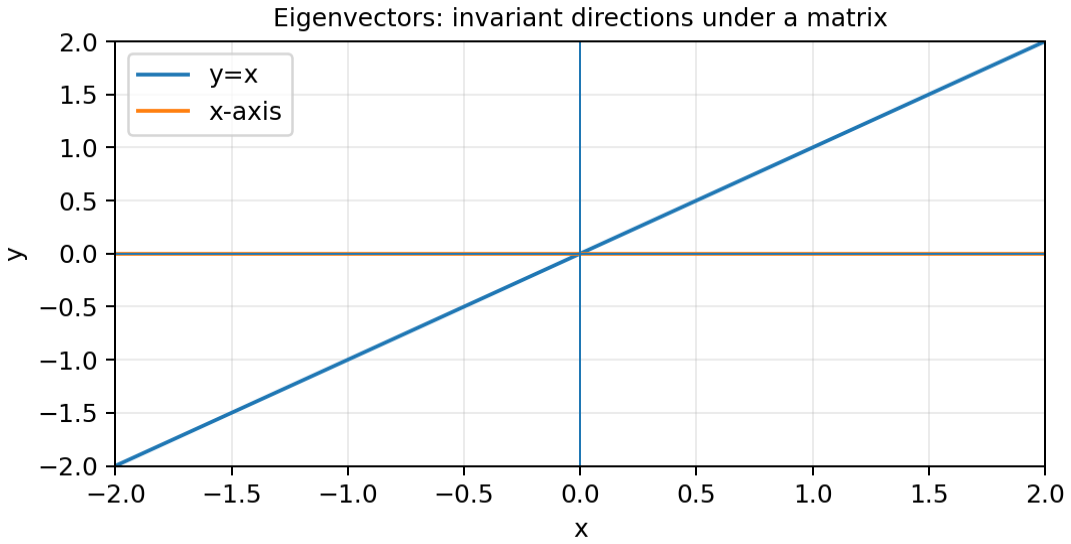
Diagonalization

An n×n matrix is diagonalizable if it has n linearly independent eigenvectors.

$A=PDP^{-1}$, where columns of P are independent eigenvectors and D contains their eigenvalues in the corresponding order.

Then $A^k=PD^kP^{-1}$ for nonnegative integer k; this can simplify repeated matrix powers.

A repeated eigenvalue does not automatically guarantee diagonalizability. Check that there are enough independent eigenvectors.



Common applications

Diagonalization can simplify matrix powers, linear dynamical systems, and coordinate transformations. Use only the applications emphasized in the lecture notes, as the syllabus instructs.

Dilation / contraction matrix reminder

In 2D, A=[[k,0],[0,k]] uniformly scales every vector by k. More generally, A=[[a,0],[0,b]] scales the coordinate axes by a and b.

Eigenvector intuition: vectors along eigen-directions stay on their own lines under A; the eigenvalue tells the signed scale factor. A negative eigenvalue reverses direction.

8. Formula sheet for fast revision

Topic	Core formula / criterion
Monotonicity	$f'>0$ increasing; $f'<0$ decreasing.
Concavity	$f''>0$ concave up; $f''<0$ concave down.
Inflection point	Concavity must change; $f''=0$ alone is not enough.
1st derivative test	$+\to-$ maximum; $-\to+$ minimum.
2nd derivative test	At $f'=0$: $f''>0$ min; $f''<0$ max; $f''=0$ inconclusive.
Two-variable test	$D=f_{xx}f_{yy}-(f_{xy})^2$; $D>0/f_{xx}>0$ min; $D>0/f_{xx}<0$ max; $D<0$ saddle.
Tangent plane	$z-f(a,b)=f_x(a,b)(x-a)+f_y(a,b)(y-b)$.
Separable ODE	$dy/h(y)=g(x)dx$; integrate.
Linear first-order ODE	$y'+Py=Q$; $\mu=e^{\int Pdx}$; $y=(\int \mu Q dx+C)/\mu$.
Exact ODE	$Mdx+Ndy=0$; exact if $M_y=N_x$.
Homogeneous ODE	$y=vx$; $y'=v+xv'$.
Constant coefficient ODE	Characteristic equation from $y=e^{rx}$.
General nonhomogeneous solution	$y=y_c+y_p$.
2×2 determinant	$ad-bc$.
Inverse	$A^{-1}=(1/\det A)\,\mathrm{adj}(A)$, if $\det A\neq 0$.
Cramer’s rule	$x_i=\det(A_i)/\det(A)$, if $\det A\neq 0$.
Consistency	$\mathrm{rank}(A)=\mathrm{rank}([A\|b])$.
Rank-nullity	$\mathrm{rank}(A)+\mathrm{nullity}(A)=$number of columns.
Eigenvalues	$\det(A-\lambda I)=0$; eigenvectors solve $(A-\lambda I)v=0$.
Diagonalization	$A=PDP^{-1}$.

9. Problem-solving checklists and common traps

Function analysis checklist

1. State the domain. 2. Compute f' and f''. 3. Find critical numbers. 4. Make sign charts for f' and f''. 5. Classify extrema and inflection points. 6. For absolute extrema on $[a,b]$, compare endpoints and interior critical points.

ODE checklist

1. Identify order and linearity. 2. Put the equation in standard form. 3. Choose the method (separable, linear, exact, homogeneous, constant-coefficient). 4. Integrate/solve carefully. 5. Apply initial conditions. 6. Check the solution by substitution.

Linear algebra checklist

1. Write $Ax=b$. 2. Row-reduce $[A|b]$. 3. Count pivots and compare ranks. 4. Identify pivot/free variables. 5. For inverse/determinant/Cramer's rule, confirm the matrix is square and $\det(A)\neq 0$ where required. 6. For diagonalization, compute eigenvalues and enough independent eigenvectors.

Frequent mistakes

Avoid	Why it matters
Calling every $f''=0$ point an inflection point	Concavity must actually change.
Using the second derivative test when $D=0$	The test is inconclusive; use another method.
Dividing by a factor in an ODE without checking zero cases	Equilibrium solutions may be lost.
Forgetting the integrating factor's constant convention	Any nonzero constant multiple works, but keep the formula consistent.
Using Cramer's rule when $\det(A)=0$	The formula requires a nonsingular coefficient matrix.
Assuming equal ranks mean unique solution	Uniqueness also requires rank(A)=number of unknowns.
Assuming repeated eigenvalues imply diagonalizable	Enough linearly independent eigenvectors are required.

Syllabus boundary

The uploaded outline explicitly says to study only the linear algebra topics mentioned in lecture notes and cites specific textbook sections/examples. This guide covers the named topic headings and standard core formulas, but it cannot reproduce the instructor's unseen class notes or guarantee that every example in the cited books is included.

Reference note

Primary syllabus-aligned references: Howard Anton, Calculus (specified sections); Ross, differential equations (specified sections); Schaum's Outline, matrix algebra. This is an original study summary, not a reproduction of those books.

End of quick review • MATH 2183 • Calculus and Linear Algebra